Q1 Solution:

a)
For strong wind period: "R1" at 80% capacity, then its power generation = 0.8 × 300 MW = 240 MW. The supply-demand curve can be constructed as follow:

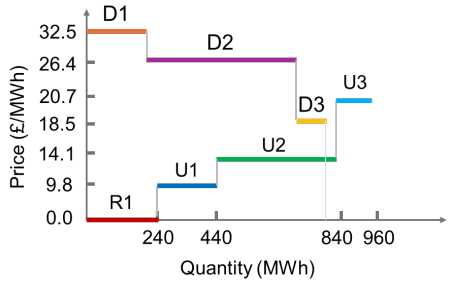


Figure Q4-1: Market supply-demand curve for strong wind period.
[1 mark]

Based on the supply-demand curve illustrated in Figure Q4-1, all the demand (D1-D3) can be fully supplied by R1, U1, and U2. As a result, the generation dispatch of the wind farm and the 3 generation units:
Generation of R1 = 240 MW
Generation of U1 = 200 MW
Generation of U2 = 360 MW
Generation of U3 = 0 MW (not dispatched)
[2 mark]

Market-clearing price = cost of marginal unit U2 = 14.1 £/MWh
[1 mark]

Generation profit of the wind farm and the 3 generation units:
Generator profit function = Dispatch × (Market price – Generation cost)
Profit of R1 = 240 MW × (14.1 – 0) £/MWh × 1 hour = £3,384
Profit of U1 = 200 MW × (14.1 – 9.8) £/MWh × 1 hour = £860
Profit of U2 = 360 MW × (14.1 – 14.1) £/MWh × 1 hour = £0
Profit of U3 = £0
[2 mark]

Demand benefit of the 3 demands:
Demand benefit function = Accepted quantity × (Bid price – Market price)
Benefit of D1 = 200 MW × (32.5 – 14.1) £/MWh × 1 hour = £3,680
Benefit of D2 = 500 MW × (28.4 – 14.1) £/MWh × 1 hour = £7,150
Benefit of D3 = 100 MW × (18.5 – 14.1) £/MWh × 1 hour = £440
[2 mark]

b)
For low wind period: "R1" at 20% capacity, then its power generation = 0.2 × 300 MW = 60 MW. The supply-demand curve can be constructed as follow:

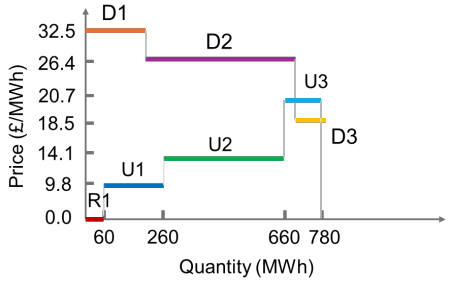


Figure Q4-2: Market supply-demand curve for perfect competition case at the off-peak period.
[1 mark]

Based on the supply-demand curve illustrated in Figure Q4-2, At price ≤ 18.5 £/MWh, supply is only 660 MW (R1+U1+U2), but demand is 800 MW (D1+D2+D3), so not enough supply. Once price reaches 20.7 £/MWh, U3 becomes available and total supply becomes 780 MW, while D3 drops out (because 18.5 < 20.7), so demand becomes 700 MW. Therefore, the market intersection is with U3 marginal.
The generation dispatch of the wind farm and the 3 generation units:
Generation of R1 = 60 MW
Generation of U1 = 200 MW
Generation of U2 = 400 MW
Generation of U3 = 40 MW
[2 mark]

Market-clearing price = cost of marginal unit U3 = 20.7 £/MWh
[1 mark]

Generation profit of the wind farm and the 3 generation units:
Generator profit function = Dispatch × (Market price – Generation cost)
Profit of R1 = 60 MW × (20.7 – 0) £/MWh × 1 hour = £1,242
Profit of U1 = 200 MW × (20.7 – 9.8) £/MWh × 1 hour = £2,180
Profit of U2 = 400 MW × (20.7 – 14.1) £/MWh × 1 hour = £2,640
Profit of U3 = 400 MW × (20.7 – 20.7) £/MWh × 1 hour = £0
[2 mark]

Demand benefit of the 3 demands:
Demand benefit function = Accepted quantity × (Bid price – Market price)
Benefit of D1 = 200 MW × (32.5 – 20.7) £/MWh × 1 hour = £2,360
Benefit of D2 = 500 MW × (28.4 – 20.7) £/MWh × 1 hour = £3,850
Benefit of D3 = 0 (not accepted)
[2 mark]

c)
Strong wind period (low price 14.1 £/MWh): Charge max
Charge power = 10 MW for 1 hour, so energy +10 MWh, storage energy: 5 + 10 = 15 MWh (within 20 kWh cap).
[1 mark]

Low wind period (high price 20.7 £/MWh): Discharge max

Discharge power = 10 MW for 1 hour, so energy −10 MWh, storage energy: 15 – 5 = 10 MWh (back to initial).
[1 mark]

Arbitrage profit = 10 MW × (20.7 – 14.1) £/MWh × 1 hour = £66
[2 mark]

Q2 solution

(a):

Zero-shot forecasting: using a pre-trained forecasting model **directly on a new time series (new wind farm / new dataset)** without task-specific training (or with minimal formatting only), producing forecasts from its learned general time-series priors.

(2 points)

Auto-regressive forecasting (AR): predict future values by conditioning on past target values, e.g.

$$\hat{y}_{t+1} = f(y_t, y_{t-1}, \dots)$$

(1 point)

Context/covariate-enhanced: predictions condition not only on past target but also on exogenous inputs (e.g., wind speed, temperature, NWP), e.g.

$$\hat{y}_{t+1} = f(y_t, y_{t-L}, x_{t:t+H})$$

Where x are covariates.

(2 points)

(b):

Two leakage examples (any two, 1 point each):

- Random train/test split (shuffling) in time series so the model trains on "future" patterns that appear in the test period.
- Normalization using full dataset statistics (mean/std computed on train+test together).
- Feature engineering that uses future info, e.g., rolling mean computed with a centered window that includes future points, or using $y_{t+1}$to build features at time t.
- Hyperparameter tuning on the test set.

Mitigation (1 point, any one):

- Use time-based split (train, validation, test in chronological (or sequential) order).
- Fit scalers only on training data and apply to val/test.
- Ensure rolling features are causal (use only past).
- Keep a strict hold-out test untouched until final evaluation.

(c):

Example: MSE or MAE is symmetric, penalizing under- and over-forecast equally, which can be misaligned when under-forecast is riskier.

(2 points)

Here, $y \geq \hat{y}$: loss = $0.9(0.40 - 0.35) = 0.9 * 0.05 = 0.045$.

$$R = 0.50 - 0.35 = 0.15 \text{ p.u.}$$

(1 point)

Pinball loss is preferred for wind power forecasting because it learns the distribution of generation uncertainty and allows explicit control of operational risk, while MSE only provides an average forecast that ignores risk asymmetry.

(1 point)

Interpretation: hold 0.15 p.u. upward reserve to cover low-wind risk at ~10th percentile.

(2 points)

(d):

$$\hat{y} = wx = 8. \text{ Error } e = \hat{y} - y = -2.$$

$$\frac{\partial \ell}{\partial w} = e \cdot x = (-2) \cdot 2 = -4.$$

$$\text{Update: } w \leftarrow w - \eta(-4) = 4 + 0.4 = 4.4.$$

(4 points for correct queation; 2 points for the correct final numerical answer.)

Q3 solution

a)

i)

1) The (Strong) Law of Large Numbers guarantees that $\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} \mathrm{Im}(s_i) = \mathrm{E}[\mathrm{Im}]$, which is used to estimate the true risk.

[1 mark]

2) The central limit theorem tells us that, for large n: $\overline{\mathrm{Im}} \xrightarrow{d} \mathcal{N}\left(\mu_{\mathrm{Im}}, \frac{\sigma^2_{\mathrm{Im}}}{n}\right)$, which is used to estimate the MC error distribution.

[1 mark]

3) Sample standard deviation is used to estimate the true standard deviation $\hat{\sigma}_{\mathrm{Im}} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(\mathrm{Im}(s_i) - \frac{1}{n}\sum_{j=1}^{n}\mathrm{Im}(s_j)\right)^2}$

[1 mark]

ii)

The coefficient of variation is defined as $\mathrm{CoV} = \mathrm{se}/\hat{r}$. The requirement is achieved when the standard error (σ') satisfies
σ'/115000=0.26%

[1 mark]

Hence, σ'= 299 and the required number of samples becomes
N'=N*(2093/299)^2=490000

[2 marks]

b)

i)

	Malicious	Accidental
Physical	Terrorism, Vandalism	Weather impacts, mechanical failure
Cyber	Hacking, Jamming of communication	Communication errors, Software bugs

[3 marks for 4 examples provided, 2 marks for 3 examples, 1 mark for 1 or 2 example(s)]

ii) The general principle is to find the behaviour/data that does not match a normal case.

[1 mark]

The normal case can be either based on physical laws or historical trend
[1 mark]

iii)

P=30, FN=10, FP=20, TN=140

TPR = 30/40 = 0.75
[1 mark]

FPR = 20/160 = 0.125
[1 mark]

c)

i) Examples include device ownership, household income, daily routine and so on

[2 marks for 3 or more examples, 1 mark for 1 or 2 examples]

The exact information that can be extracted depends on the resolution data
[1 mark]

ii) List three from the below:
Pseudonymisation: Replacing identifiable features with consistent unique identifiers.
Aggregation: Aggregating consumption data across multiple periods of time (temporal) or households(spatial).
Differential Privacy: Carefully tuned noise addition to 'hide' individual contributions.
Homomorphic Encryption: Perform arithmetic operations (e.g., addition, multiplication) on encrypted data without having to first decrypt it.
User Demand Shaping: Altering actual consumption patterns using flexible assets to hide appliance characteristics.
Federated Learning: Distributed learning technique in which model parameters are shared but raw data is kept locally.
[2 marks for three or more examples, 1 mark for 1-2 examples]

The answer needs to cover the pros/cons of the listed three options based on the impact on utility of the smart meter data, the complexity of the implementation, and the compatibility with existing infrastructures.
[2 marks for three or more examples, 1 mark for 1-2 examples]

Q4 Solution

(a):

DC branch-flow model:

$$P_{\text{br}} = \text{diag}(b)A\theta \Rightarrow H = \text{diag}(b)A.$$

$$\text{diag}(b) = \text{diag}(2.5,10,5,5,2)$$

$$H = \begin{bmatrix} 2.5 & -2.5 & 0 & 0 \\ 10 & 0 & 0 & -10 \\ 0 & 5 & -5 & 0 \\ 0 & 5 & 0 & -5 \\ 0 & 0 & 2 & -2 \end{bmatrix}$$

(2 points)

With bus 1 slack ($\theta_1 = 0$), states are $x = [\theta_2, \theta_3, \theta_4]^{\top}$. Remove column 1:

$$H_{\text{red}} = \begin{bmatrix} -2.5 & 0 & 0 \\ 0 & 0 & -10 \\ 5 & -5 & 0 \\ 5 & 0 & -5 \\ 0 & 2 & -2 \end{bmatrix}$$

The rank is 3 (full column rank) and the unknown state dimension is 3. Therefore the system is observable.

(2 points)

(b):

Bus injections (MW) from branch flows:

$$P_{\text{inj}} = A^{\top} P_{\text{br}}$$

Using $P_{\text{br}} = [50, -500, -50, 150, 80]^{\top}$MW:

$$P_1 = P_{12} + P_{14} = 50 + 500 = 550\text{MW}$$
$$P_2 = -P_{12} + P_{23} + P_{24} = -50 - 50 + 150 = 50\text{MW}$$
$$P_3 = -P_{23} + P_{34} = 50 + 80 = 130\text{MW}$$
$$P_4 = -P_{14} - P_{24} - P_{34} = -500 - 150 - 80 = -730\text{MW}$$

(2 points)

The most likely load bus is Bus 4 as it has the largest negative injection in magnitude.

(1 point)

(c):

Given $S_{\text{base}} = 100\text{MVA}$ and slack bus 1: $\theta_1 = 0$.

DC branch-flow equations:

$$P_{ij} = b_{ij}(\theta_i - \theta_j).$$

The state of each bus can be computed as follows.

$$P_{12} = b_{12}(\theta_1 - \theta_2) = 2.5(0 - \theta_2) = 0.5 \Rightarrow \theta_2 = -0.2\text{rad}.$$

$$P_{14} = b_{14}(\theta_1 - \theta_4) = 10(0 - \theta_4) = 5.0 \Rightarrow \theta_4 = -0.5\text{rad}.$$

$$P_{23} = b_{23}(\theta_2 - \theta_3) = -0.5(-0.2 - \theta_3) = -0.5 \Rightarrow \theta_3 = -0.1\text{rad}.$$

(2 points)

Compute the predicted values for the remaining measured lines:

$$\hat{P}_{24} = 5(\theta_2 - \theta_4) = 5(-0.2 - (-0.5)) = 5(0.3) = 1.5 \text{ p.u.} = 150\text{MW}$$

$$\hat{P}_{34} = 2(\theta_3 - \theta_4) = 2(-0.1 - (-0.5)) = 2(0.4) = 0.8 \text{ p.u.} = 80\text{MW}$$

As the measurements match exactly, $r = 0$ (under noiseless condition).

(1 point)

(d):

No. A classic residual-based BDD flags attacks that produce a nonzero residual $r = z - H\hat{x}$. A stealth FDIA must satisfy

$$a \in \text{Im}(H) \Leftrightarrow a = Hc$$

for some vector c. Changing only one measurement component (only P_{14}) corresponds to an attack vector with one nonzero entry, which is not (generically) in the column space of H. Hence it will typically be detected.

(2 point)

If the attacker only changes θ_4, take

$$c = \begin{bmatrix} 0 \\ 0 \\ \Delta\theta_4 \end{bmatrix}$$

By the measurement matrix and converting to MW,

$$a^{MW} = 100a^{pu} = \Delta\theta_4 \begin{bmatrix} 0 \\ -1000 \\ 0 \\ -500 \\ -200 \end{bmatrix}$$

Therefore, from the 2nd entry:

$$-1000\Delta\theta_4 = +300 \Rightarrow \Delta\theta_4 = -0.3\text{rad}$$

Therefore the stealth attack vector that bypasses BDD (noiseless) is:

$$a^{MW} = \begin{bmatrix} 0 \\ +300 \\ 0 \\ +150 \\ +60 \end{bmatrix} \text{MW}$$

(3 points)

(e):

Under normal conditions, $J \sim \chi^2(\text{DoF})$ approximately. (Or, it follows chi-square distribution.)

(1 point)

DoF=m−n=5−3=2.

(1 point)

(f):

Moving Target Defence (MTD) in power-system cyber-security means deliberately and periodically perturbing system parameters (commonly line reactances via FACTS, switching, tap changes, etc.) so that the attacker's assumed measurement model H becomes inaccurate. If the attacker designs a *stealth* FDIA using the old model (e.g. $a = Hc$), after MTD the true model is $H' \neq H$, so the injected data no longer lies in $\text{Im}(H')$and produces a nonzero residual, making BDD/estimation detect it.

(1 point)

From (d), the attacker changes only θ_4, so their stealth attack is

$$a = H_{\text{red}}c, c = [0,0,\Delta\theta_4]^{\top},$$

i.e., a is proportional to the θ_4 column of H_{red}. In this network, the θ_4 column depends only on branches incident to bus 4: line 1-4, 2-4, 3-4. Changing the reactance of any one of these will change H in the directions that the attacker relies on.

(2 points; if not all three lines/reactances are answer, only 1 point is given).